

SIGNIFNT = 8

Tuesday, November 11, 2025 9:49 PM

Q1. $m = \text{Flow Rate} \times \text{Density}$

$$m_{\text{H}_2\text{O}} = 1.4707 \times 62$$

$$= 91.17 \text{ lb/min}$$

$$Q = m C_p \Delta T$$

$$Q_{\text{water}} = 91.17 \times 1 \times 50$$

$$= 4558.5 \text{ BTU/min}$$

$$\Delta T = 250 - 120 = 130^\circ\text{F}$$

$$Q_{\text{oil}} = 38.1 \times 0.55 \times 130$$

$$= 2744.7 \text{ BTU/min}$$

$$Q = U_o \times A \times \Delta T_{\text{lm}}$$

$$\Delta T_{\text{lm}} = \frac{(\Delta T_1 - \Delta T_2)}{\ln\left(\frac{\Delta T_1}{\Delta T_2}\right)}$$

$$= \frac{130 - 0}{\ln\left(\frac{130}{0}\right)} = 130^\circ\text{F}$$

For co current flow

$$\Delta T_1 = 250 - 70 = 180^\circ\text{F}$$

$$\Delta T_2 = 120 - 120 = 0^\circ\text{F}$$

$$A = \frac{Q}{U_o \times \Delta T_{\text{lm}}}$$

$$A_{\text{co}} = \frac{2744.7}{110 \times 130} = \frac{2744.7}{14300} = 0.192 \text{ ft}^2$$

$$A_{\text{co}} = 1.5 \times 0.1387 = 0.208 \text{ ft}^2$$

$$A_{\text{section}} = 3.14 \times 0.125 \times 12$$

$$= 4.71 \text{ ft}^2$$

Co current Co current

$$\text{No. of Sections} = \frac{0.192}{4.71} = 0.04$$

Co current

$$\text{No. of Sections} = \frac{0.208}{4.71} = 0.044$$

Q2)

$$Bi = \frac{h L_c}{k} = \frac{h \cdot d}{2k} = \frac{25 \times 0.01}{2 \times 40}$$

$$= \frac{25 \times 0.005}{40} = 0.003125$$

LCH valid

$$\frac{T(t) - T_\infty}{T_i - T_\infty} = \exp\left(-\frac{h A}{\rho C V} t\right)$$

$$A = 4\pi \left(\frac{d}{2}\right)^2 = 4\pi \left(\frac{0.01}{2}\right)^2 = 7.854 \times 10^{-4} \text{ m}^2$$

$$V = \frac{4}{3} \pi (0.005)^3 = 5.236 \times 10^{-7} \text{ m}^3$$

$$t = \frac{\rho C V}{h A} \ln\left(\frac{T(t) - T_\infty}{T_i - T_\infty}\right)$$

$$t = \frac{7800 \times 600 \times 5.2 \times 10^{-4}}{25 \times 7.854 \times 10^{-4}} \ln\left(\frac{450 - 325}{1150 - 325}\right)$$

$$\therefore t = 4.81 \text{ mins}$$

Q6)

$$Q = \frac{k A \Delta T}{L}$$

$$L = \frac{2.0}{20} = 0.1 \text{ inches}$$

$$k_{\text{copper}} = 385 \times 0.855 = 21.3675 \text{ W/in } ^\circ\text{F}$$

$$k_{\text{glass}} = 1.4 \times 0.555 = 0.8777 \text{ W/in } ^\circ\text{F}$$

$$R_{\text{parallel}} = 10 \left(\frac{L_{\text{layer}}}{k_{\text{copper}} \times A} + \frac{L_{\text{layer}}}{k_{\text{glass}} \times A} \right)$$

$$R_{\text{parallel}} = 10 \left(\frac{0.1}{21.3675} + \frac{0.1}{0.8777} \right) = 12.92$$

$$\underline{R_{\text{normal}} = 12.92}$$

$$\therefore R_{\text{ratio}} = 1:1$$

will heat equally